



ENVISAGE: An image captioning dataset and conformal uncertainty modeling framework to assist visually impaired individuals

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ABSTRACT

Image captioning has become a key task in vision-language research, bridging visual understanding and natural language generation. However, existing image captioning datasets and models are primarily optimized for sighted individuals, often failing to address the needs of visually impaired individuals who rely on captions for scene understanding and navigation. To address this gap, we introduce ENVISAGE, a large-scale image captioning dataset curated with input from visually impaired individuals. It contains 53,452 images and 267,260 accessibility-focused captions that emphasize spatial relations and object affordances. We benchmark both baseline (*Show and Tell*, *Show, Attend and Tell*) and state-of-the-art (*BLIP* and *BLIP-2*) captioning models on ENVISAGE under zero-shot and fine-tuned settings. Beyond the dataset contribution, we present the first framework for conformal uncertainty estimation in image captioning, providing finite-sample coverage guarantees that improve confidence in caption quality. Results demonstrate that conformal predictions achieve reliable coverage guarantees across diverse image categories. Among all evaluated models, the fine-tuned *BLIP-2 OPT6.7B* achieves the best overall performance, with a final score of 0.4362 across the image captioning metrics. Furthermore, uncertainty-aware evaluation shows that captions in the low-uncertainty band attain the highest quality, achieving a final score of 0.4476, confirming that the proposed conformal framework produces calibrated uncertainty estimates that align with caption reliability. ENVISAGE and our uncertainty-aware framework establish a robust and accessible foundation for image captioning systems.

1. Introduction

Image captioning-the task of automatically generating descriptive natural language sentences from images-has made substantial progress over the past decade [1]. This multimodal task, requiring both computer vision and natural language processing capabilities, has emerged as a critical benchmark for evaluating machine understanding of visual scenes [2]. The ability to translate visual information into coherent textual descriptions plays a crucial role in real-world applications such as image retrieval [3], content moderation [4], and, most importantly, accessibility for visually impaired individuals [5]. Limited access to visual information introduces significant challenges in daily activities such as navigation, object recognition, and reading, often increasing dependence on others among visually impaired individuals [6]. Assistive technologies that convert visual content into accessible modalities are therefore essential for enhancing independence and quality of life [7]. Initial efforts in image captioning relied heavily on template-based or retrieval-based methods [8], which utilized hand-crafted features or retrieved similar captions based on image similarity.

While these methods demonstrated the feasibility of the task, they were limited by static vocabularies and poor generalization to novel scenes or object compositions. A paradigm shift occurred with the introduction of deep learning approaches, where convolutional neural networks (CNNs) learned visual representations directly from images and recurrent neural networks (RNNs) produced captions directly from data, addressing the limitations of hand-crafted or retrieval-based approaches [9]. The encoder-decoder framework, proposed by Vinyals et al. [10], combined CNNs as image encoders with RNNs as text decoders, enabling end-to-end training from pixels to words. This architecture allowed visual features to be directly mapped into linguistic sequences, resulting in improvements in the fluency, grammatical correctness, and relevance of generated captions. Further improvements were introduced with attention mechanisms [11], allowing models to dynamically focus on salient regions of an image during the generation of each word. Beyond performance gains, these mechanisms also enhanced interpretability by providing alignment between image regions and textual tokens. This foundation led to more sophisticated

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architectures such as the transformer [12], which replaced recurrence with self-attention and enabled parallel computation and deeper representation learning. Meanwhile, the emergence of large-scale vision-language pre-training has fundamentally advanced multimodal learning. Models such as OSCAR [13], CLIP [14], BLIP [15], and Flamingo [16] leverage massive corpora of image-text pairs to learn generalizable representations. These models enable zero-shot captioning, improved robustness to out-of-domain content, and enhanced ability to describe new objects or unfamiliar scenarios [17], achieving state-of-the-art performance across multiple vision-language benchmarks. However, despite these significant advances, applying these systems to accessibility remains challenging. Existing captioning systems are mostly built on general-purpose datasets such as MS COCO [18], Flickr8k [19], Flickr30k [20], and Visual Genome [21]. While these datasets have driven significant progress, they lack accessibility considerations. Their captions generally provide brief descriptions without spatial reasoning, textual content, or subtle semantic details that the visually impaired require for environmental navigation. Datasets such as VizWiz-Captions [22] and TextCaps [23] have been introduced to address this gap. VizWiz-Captions, sourced directly from visually impaired individuals, captures real-world image quality issues (e.g., blur, occlusion) and authentic usage scenarios. TextCaps, meanwhile, focuses on incorporating optical character recognition to enable reading of signs, labels, and documents in natural scenes. Despite their contributions, these datasets have limitations: VizWiz-Captions are often short or imprecise due to open-ended crowd-sourced annotations, while TextCaps, centered on scene text, lacks broader contextual understanding or spatial relationships between visual entities. To address these limitations, we introduce ENVISAGE, the first accessibility-focused image captioning dataset co-designed with visually impaired individuals. ENVISAGE consists of 53,452 images collected in collaboration with blind and low-vision individuals, each annotated with five captions that explicitly emphasize scene layout, spatial relationships between objects, and task-relevant visual cues. Unlike prior datasets, captions in ENVISAGE were written under the instruction to simulate a “visual assistant” providing spoken descriptions to a user, prioritizing clarity, spatial awareness, and functional relevance. Beyond these detailed captions, each image is annotated with multi-label tags derived from MS COCO supercategories, enabling multi-task modeling and rich semantic understanding. Building accessibility-focused captioning systems using ENVISAGE requires addressing a fundamental challenge. For visually impaired individuals, automated descriptions serve as a primary means of perceiving and interpreting visual information, making reliability equally important to accuracy. As ENVISAGE emphasizes realistic, user-centered imagery and spoken-style annotations, it highlights the limitations of conventional captioning models that produce deterministic outputs without accounting for uncertainty. This motivates the development of uncertainty-aware captioning approaches, wherein models not only generate semantically rich and contextually grounded descriptions but also quantify their confidence in each part of the output. These complementary advances—detailed accessibility-focused annotations paired with principled uncertainty quantification—enable more reliable and practical deployment of vision-language systems in real-world assistive scenarios. Fei et al. [24] introduce uncertainty-aware image captioning through an insertion-based strategy that ranks candidate words by their uncertainty, first generating high-confidence keywords and subsequently inserting uncertain auxiliary terms. This non-autoregressive approach improves decoding speed and yields better CIDEr scores, demonstrating that uncertainty cues can effectively guide generation. Bayesian Self-Critical Sequence Training [25] extends reinforcement-learning-based captioning with Monte Carlo dropout to estimate epistemic uncertainty (i.e. uncertainty caused by the lack of knowledge). By averaging multiple sampled captions, the method derives a more stable baseline for policy gradients and produces uncertainty estimates that correlate with caption quality. Petryk et al. [26] introduce Token-Level Confidence (TLC), a simple yet effective confidence metric

that aggregates token probabilities from a captioning model. Their results demonstrate that TLC not only improves fine-grained correctness on compositional tasks but also reduces object hallucinations when used to rerank candidate captions. Zheng et al. [27] incorporate subjective logic to model vacuity and dissonance in medical image captioning. Their interactive framework predicts keywords with associated uncertainties, allows users to validate uncertain terms, and refines captions based on this feedback, reducing errors in domain-specific scenarios. Yu et al. [28] propose a quality-agnostic captioning approach for visually impaired individuals that augments data with synthetic noise and employs a dual-branch architecture to explicitly handle noisy and clean visual input. They calibrate model confidence across different image qualities, leading to more robust predictions and better alignment between confidence scores and correctness. Although these methods demonstrate the value of uncertainty modeling, they lack statistically rigorous calibration guarantees and interpretable uncertainty measures suitable for reliable captions. To address this limitation, we propose a conformal prediction-based image captioning framework that provides rigorous, instance-level uncertainty quantification within a semiparametric setting. Following recent advances in image captioning, we adopt a parametric model, as is standard in the literature, due to its ability to learn complex, high-dimensional mappings from images to natural language through a fixed set of parameters. However, such models do not inherently provide calibrated uncertainty estimates. To address this, we introduce a nonparametric conformal calibration layer, which estimates empirical quantiles over token-level negative log-likelihood (NLL) scores using a held-out calibration set, without making distributional assumptions. Uncertainty levels (low, medium, high, very high) are derived from this nonparametric layer, providing finite-sample marginal coverage guarantees under minimal assumptions. As a result, the model outputs captions accompanied by calibrated, interpretable uncertainty estimates at both the word and sequence levels, improving reliability and supporting informed decision-making for visually impaired users. Unlike existing approaches that rely on heuristic confidence measures, our framework provides distribution-free, finite-sample calibrated uncertainty estimates for image captioning, enabling reliable and interpretable predictions in assistive settings.

In summary, our main contributions are:

- **ENVISAGE Dataset:** A large-scale, accessibility-focused image captioning dataset collected in collaboration with visually impaired individuals, emphasizing spatial relationships, contextual scene understanding, and real-world assistive scenarios.
- **Conformal Prediction-Based Captioning Framework:** A novel uncertainty-aware captioning approach that integrates conformal prediction to provide statistically rigorous uncertainty estimates at both word and sequence levels, with predictions categorized into discrete uncertainty levels (low, medium, high, very high) to quantify caption reliability.
- **Comprehensive Evaluation:** Extensive benchmarking of state-of-the-art captioning models on ENVISAGE, revealing limitations of existing approaches in assistive contexts and demonstrating that conformal prediction enables more reliable and informative descriptions for visually impaired individuals.

The remainder of this paper is organized as follows. Section 2 describes the ENVISAGE dataset, including its collection process, annotation protocols, and statistical analysis. Section 3 outlines the baseline models and introduces our proposed conformal prediction-based uncertainty-aware captioning framework. Section 4 presents quantitative results and qualitative analyses. Finally, Section 5 summarizes the main findings, discusses limitations, and outlines directions for future research.

For clarity, all variables and parameters appearing in the proposed framework are defined in Table 1. We refer to this table throughout the section to avoid repetition.

Table 1
General notation used in the proposed framework.

Symbol	Description
I	Input image
$\hat{y}_{1:T}$	Generated caption sequence of length T
$\hat{y}_t$	Token generated at position t
$\hat{y}_{<t}$	Previously generated tokens before position t
T	Number of tokens in a caption
θ	Parameters of the captioning model
$p_\theta(\cdot)$	Conditional probability defined by the model
α_t	Token-level nonconformity score
$\tilde{\alpha}$	Caption-level nonconformity score
$\mathcal{A}_{\text{cal}}$	Set of token-level calibration scores
$\mathcal{B}_{\text{cal}}$	Set of caption-level calibration scores
N	Number of calibration tokens
M	Number of calibration samples (captions)
ϵ	Miscoverage level ($1 - \epsilon$ is target coverage)
q_k	Coverage quantile (e.g., 0.68, 0.95, 0.997)
$\hat{\tau}_{q_k}$	Token-level conformal threshold at level q_k
$\tilde{\tau}_{q_k}$	Caption-level conformal threshold at level q_k
$\mathcal{D}_{\text{cal}}$	Calibration set

2. The ENVISAGE dataset

This section introduces ENVISAGE, a large-scale image captioning dataset co-designed with visually impaired individuals to address real-world accessibility needs.¹

2.1. Dataset creation

The creation of the ENVISAGE dataset employed clearly defined protocols (Section 2.1.1) to ensure ethical data collection, high linguistic quality, and real-world relevance for visually impaired individuals. These protocols addressed both the practical challenges faced by visually impaired individuals in daily life and the technical requirements of assistive image captioning systems.

2.1.1. Image collection protocol

Images were collected from visually impaired individuals through voluntary participation in collaboration with the Association for the Protection of the Blind in İzmir, Türkiye. Participants used their personal smartphones to capture scenes reflecting their everyday environments and experiences. All participants provided informed consent, and the study received ethical approval from the Social Research Ethics Committee of İzmir Kâtip Çelebi University (Decision No: 2021/15-05, dated August 5, 2021).

To emulate real-world visual conditions encountered by visually impaired individuals, participants were encouraged to prioritize naturalistic scenes over technical perfection. Images with slight focus issues, poor lighting, or partial occlusions were accepted, provided they met basic content and ethical standards. The following procedural guidelines ensured privacy, quality, and diversity:

- Capture contextual scenes rather than isolated object close-ups.
- Avoid repetitive images from identical viewpoints.
- Exclude personally identifiable information (IDs, bank cards, documents with names/dates).
- Remove camera metadata (brand, model, timestamps).

Images were captured primarily by visually impaired individuals residing in İzmir, Türkiye. The initial batch was collected on August 24, 2021. All submissions were manually reviewed by the authors to ensure compliance with the established standards before inclusion in the dataset.

Table 2
Summary statistics of the ENVISAGE dataset.

General statistics	Value	Vocabulary statistics	Value
Total images	53,452	Vocabulary size	15,809
Total captions	267,260	Caption uniqueness	91.95%
Average caption length	9.01 ± 2.10	Type-Token Ratio	0.007
Caption length range	[6, 15]	Flesch–Kincaid Grade Level	3.74

2.1.2. Annotation protocol

Each image was annotated with five English captions written by native speakers following accessibility-focused annotation guidelines. Annotators were instructed to:

- **Content and style:** Describe all visually important elements clearly and objectively, as if explaining the scene to the visually impaired over the phone.
- **Length and structure:** Provide a single, descriptive sentence of 6 to 15 words without generic openings like “There is...” or “This is...”.
- **Objectivity:** Avoid speculation about unseen actions, thoughts, or events beyond the visible content.
- **Exclusions:** Omit proper nouns, punctuation, or camera angle (e.g., blurry, tilted).
- **Text handling:** When relevant text appears, summarize its meaning rather than quoting directly.

Annotations were conducted by the American Cultural Association, ensuring high linguistic quality. The authors conducted quality assurance reviews, examining captions for accuracy, clarity, and adherence to the established guidelines. Captions that did not meet these standards were revised based on detailed feedback. This review process ensured that the final captions were reliable, and aligned with the accessibility goals of the dataset. A recurring challenge during annotation was that some images lacked sufficient contextual information to fully satisfy the established guidelines. In such cases, annotators were instructed to provide the most accurate and concise description possible within the defined constraints.

2.2. Dataset statistics

This section summarizes the statistics of the ENVISAGE dataset, focusing on caption properties, linguistic patterns, and semantic alignment with images.

2.2.1. Caption length, vocabulary, and readability

Table 2 summarizes the ENVISAGE dataset. With 53,452 images and 267,260 captions (5 per image), the dataset achieves a balance between scale and annotation density, enabling effective training and evaluation of both deep learning models and few-shot techniques.

The captions have a length distribution ranging from 6 and 15 words, with an average of 9.01 ± 2.10 words. This controlled yet variable range indicates that annotators consistently produced short, informative descriptions. The average caption length reflects a balance between detail and usability, a key design feature for real-time assistive systems. Longer captions may provide more information but risk overloading audio output or mobile processing capabilities; shorter captions, conversely, may lack essential context. The narrow standard deviation highlights the consistency of the annotation process, while the upper limit of 15 words ensures conciseness for screen-reader compatibility. The lower bound of 6 words guarantees sufficient descriptiveness, avoiding trivial or overly vague captions.

As shown in Table 2, the vocabulary comprises 15,809 unique words - considerably diverse for a constrained-length captioning dataset. This breadth supports generalization and fine-grained distinctions in model predictions. The caption uniqueness rate of 91.95% indicates that annotators used diverse linguistic constructions even for visually similar

¹ The dataset is available at: https://github.com/cayl-crypto/envisage_data_set.

Table 3

Top 10 most frequent words, bigrams, and trigrams in captions.

Word	Freq.	Bigram	Freq.	Trigram	Freq.
front	31,927	on the	54,890	in front of	21,292
trees	26,162	of the	38,750	front of the	10,338
side	24,050	in front	23,888	is on the	9298
behind	17,462	in the	23,150	on the right	8177
building	17,102	front of	21,878	are on the	7669
people	17,006	of a	15,613	on the left	7630
right	16,302	on a	15,039	side of the	6421
standing	16,200	is on	11,873	front of a	5308
road	16,187	are on	10,227	the right side	3386
background	15,846	the right	9895	of the view	3381

scenes, which helps mitigate model biases and makes the dataset well-suited for tasks like paraphrase generation or style transfer. While the Type-Token Ratio [29] of 0.007 appears low in absolute terms, it is as expected for a large corpus. This indicates that a small set of high-frequency tokens (e.g., “on”, “the”, “in”) dominates, but the long tail of rare terms remains substantial. This is ideal for training robust models that can leverage both common and rare linguistic features. The Flesch–Kincaid Grade Level [30] of 3.74 indicates that the captions use simple sentence structures and accessible vocabulary, resulting in text that is easy to read and process. This metric quantifies linguistic simplicity by evaluating factors such as sentence length and word complexity. We highlight this measure because it provides an objective assessment of cognitive accessibility: captions written at this level are concise, clear, and suitable for rapid comprehension. This is essential not only for end users who may rely on screen readers, but also for downstream applications where simplified language improves processing speed and reduces the risk of model hallucinations.

2.2.2. Linguistic structure and spatial relationships

Table 3 lists the most common non-stopwords. Words such as “front”, “side”, “behind”, “standing”, and “road” indicate that captions emphasize spatial arrangements and relative positioning. This reflects an emphasis on describing object relationships, which is critical information for visually impaired individuals. Nouns like “trees”, “people”, “building”, and “road” also dominate the list, suggesting that scenes predominantly depict structured outdoor environments with recognizable landmarks.

Table 3 shows that the most common bigrams include “on the”, “of the”, “in front”, and “in the”. These prepositional phrases express object-location relationships, e.g., “a man is standing in front of a building”. The prominence of these structures indicates that captions establish foreground-background and object-to-object relations, which are crucial for scene understanding. Frequent subject-verb constructs such as “people are”, “are on”, and “is on” indicate that captions often describe active scenes rather than static object listings. This reflects a balanced mix of noun-centric and action-oriented descriptions, both of which are valuable for assisting visually impaired individuals.

The trigrams in Table 3 reinforce spatial narrative patterns. Phrases like “in front of”, “on the right”, “on the left”, and “side of the” illustrate an emergent grammar of spatial layout. These structures are not only common but also semantically vital, they help the visually impaired construct mental maps of the visual scenes. Moreover, the trigram “cars are parked” suggests repetitive but realistic urban content, likely from sidewalks or parking lots, visually complex scenes particularly relevant for mobility assistance. The trigram “next to the” further confirms that relative object positioning is a central theme of this dataset.

2.2.3. Alignment between linguistic patterns and visual content

There is strong alignment between the linguistic patterns and its visual tag distribution of the dataset. For example:

- The frequent use of spatial terms such as “front”, “in front of”, and “on the left” correlates with the prevalence of multi-object outdoor scenes where such descriptions are meaningful.
- The dominance of nouns such as “people”, “trees”, and “buildings” corresponds directly to the most common image tags.
- Sentence structures emphasizing spatial relations align with the goal of enhancing accessibility, these phrases help the visually impaired build conceptual understanding of visual layouts.

Image tag distribution in the dataset is as follows:

- Outdoor: 76%
- Vehicle: 24%
- Person: 21%
- Indoor: 18%
- Furniture: 8%
- Animal: 6%
- Food: 5%
- Kitchen: 2%
- Accessory: 2%
- Electronic: 2%
- Appliance: 1%
- Sports: 1%

This distribution reflects real-world scenarios commonly encountered by visually impaired individuals, with strong emphasis on outdoor environments and scenarios involving people. While categories like “Appliance”, “Accessory” and “Sports” are relatively rare, their inclusion ensures coverage of a broad range of visual environments. For downstream modeling, this imbalance can be addressed through category-aware learning strategies, such as weighted sampling or class-balanced loss functions.

The ENVISAGE dataset demonstrates rich lexical diversity, robust spatial semantics, and strong alignment between linguistic patterns and visual content, making it well-suited for training and evaluating vision-language models for accessibility applications.

2.3. Data splitting

To enable fair benchmarking and reproducible experiments, the ENVISAGE dataset was split into 10 different training/validation/test configurations, named `split_0` through `split_9`. Each split was created using stratified sampling based on multi-label tag distributions. Stratification ensures that each split maintains consistent semantic composition across scene categories. Split proportions are fixed at 70% for training, 10% for validation, and 20% for testing.

2.3.1. Split selection and performance analysis

While all 10 splits maintain consistent tag distributions, we select the optimal split based on the quality of GPT-4o-generated captions. For each split, we first compute eight standard image captioning metrics, namely BLEU-1 through BLEU-4 [31], METEOR [32], ROUGE-L [33], CIDEr [34], and SPICE [35], on the training, validation, and test sets, obtaining a score for each subset. For each evaluation metric, we then aggregate the training, validation, and test scores using the harmonic mean, encouraging balanced performance across data splits. Using these aggregated metrics, we compute a final score as defined in Eq. (1), by first averaging BLEU-1 to BLEU-4 into a single BLEU score and then taking the arithmetic mean of this score with METEOR, ROUGE-L, CIDEr, and SPICE. This final aggregate score is used to evaluate all splits and select the optimal benchmark split.

$$\text{Final Score} = \frac{1}{5} \left(\left(\frac{1}{4} \sum_{i=1}^4 \text{BLEU}_i \right) + \text{METEOR} + \text{ROUGE-L} + \text{CIDEr} + \text{SPICE} \right) \quad (1)$$

Table 4

Harmonic mean scores across the 10 splits. Highest values in each column are shown in bold. `split_4` achieves the highest harmonic mean score.

Split	H-BLEU-4	H-CIDEr	H-SPICE	H-Final Score
0	0.0445	0.3179	0.1362	0.2134
1	0.0453	0.3188	0.1367	0.2141
2	0.0455	0.3201	0.1372	0.2144
3	0.0447	0.3183	0.1369	0.2135
4	0.0452	0.3210	0.1375	0.2147
5	0.0434	0.3171	0.1365	0.2129
6	0.0445	0.3197	0.1371	0.2137
7	0.0450	0.3195	0.1376	0.2144
8	0.0452	0.3199	0.1373	0.2142
9	0.0454	0.3211	0.1368	0.2144

Table 5

Captioning performance on `split_4` across training, validation, and test sets.

Subset	BLEU-4	CIDEr	SPICE	Final Score
Train	0.0450	0.3160	0.1365	0.2132
Validation	0.0465	0.3285	0.1395	0.2175
Test	0.0443	0.3186	0.1366	0.2135
Harmonic mean	0.0452	0.3210	0.1375	0.2147

where BLEU_i denotes the BLEU score with i -gram precision, for $i \in \{1, 2, 3, 4\}$.

As shown in Table 4, the highest harmonic mean final score of **0.2147** was achieved by `split_4`. It outperformed other splits on most metrics while achieving competitive scores on the rest. While `split_7` had the highest SPICE score, its overall harmonic mean was lower. Therefore, `split_4` was selected as the official benchmark split.

Table 5 shows captioning performance for `split_4`, broken down by subset. Test performance (e.g., CIDEr: 0.3186, SPICE: 0.1366) aligns closely with training and validation scores, indicating strong generalization without overfitting.

3. Baselines and proposed framework

In this section, we first summarize the baseline captioning models evaluated on our dataset, then introduce our conformal-based uncertainty-aware framework. After evaluating multiple candidates, we integrate the best-performing baseline into the proposed framework to enable reliable, uncertainty-aware caption generation. Finally, we present the theoretical analysis of the proposed framework.

3.1. Baseline models

We evaluated three categories of image captioning models on the ENVISAGE dataset: classical encoder-decoder architectures, transformer-based captioning models, and large pretrained vision-language models. These baselines serve as strong benchmarks and allow us to isolate the impact of our proposed uncertainty-aware framework.

Classical Models: We include *Show and Tell* [10] and *Show, Attend and Tell* [11], which follow the standard CNN-RNN paradigm. *Show and Tell* employs a ResNet-based encoder with a Long-Short Term Memory decoder, while *Show, Attend and Tell* extends this architecture by incorporating spatial attention over convolutional feature maps.

Transformer-Based Models: To incorporate modern sequence modeling, we evaluate ViT-GPT2 [36] and CvT-GPT2 [37] captioners, where a vision transformer (ViT or CvT) extracts global visual embeddings and a GPT-2 decoder autoregressively generates captions.

Pretrained Vision-Language Models: We evaluate *BLIP* [15] and *BLIP-2* [38] variants, as well as *GPT-4o* [39], which represent state-of-the-art pretrained captioning paradigms. Specifically, we evaluate *BLIP* Base and *BLIP* Large (both pretrained on COCO) alongside three *BLIP-2*

variants-*BLIP-2 OPT2.7B*, *BLIP-2 OPT6.7B*, and *BLIP-2 T5*. These models combine vision encoders with language models of different scales, enabling systematic evaluation of how model scale affects caption quality. All *BLIP* and *BLIP-2* variants are evaluated in both zero-shot and fine-tuned settings to assess their adaptability to ENVISAGE. Since *GPT-4o* is only available for zero-shot use, we report its out-of-the-box performance as a reference for generalization.

3.2. Proposed framework

We propose a conformal-based uncertainty-aware image captioning framework that augments the best-performing baseline model, *BLIP-2 OPT6.7B* [38], with calibrated uncertainty estimates at both the token and sentence levels. While conventional captioning models produce a single deterministic output, they provide no insight into the reliability of each generated word or the caption as a whole. To address this limitation, our approach quantifies uncertainty using conformal prediction [40], enabling interpretable caption generation. Given an input image I and a generated caption $\hat{y}_{1:T}$, the captioning model produces token probabilities

$$p_\theta(\hat{y}_t | \hat{y}_{<t}, I), \quad t = 1, \dots, T. \quad (2)$$

We define the token-level nonconformity score using the negative log-likelihood:

$$\alpha_t = -\log p_\theta(\hat{y}_t | \hat{y}_{<t}, I), \quad (3)$$

where larger values indicate higher uncertainty.

Using a held-out calibration set $\mathcal{D}_{\text{cal}}$, we construct the set of token-level nonconformity scores:

$$\mathcal{A}_{\text{cal}} = \{\alpha_t^{(i)} : i = 1, \dots, M, t = 1, \dots, T_i\}. \quad (4)$$

For a desired coverage level $1 - \epsilon$, we compute the empirical conformal threshold

$$\hat{\tau}_{1-\epsilon} = \text{Quantile}_{1-\epsilon}(\mathcal{A}_{\text{cal}}), \quad (5)$$

which yields the finite-sample guarantee

$$\mathbb{P}(\alpha_{n+1} \leq \hat{\tau}_{1-\epsilon}) \geq 1 - \epsilon. \quad (6)$$

During inference, each token is labeled as uncertain if

$$\mathbb{I}(\hat{y}_t \text{ is uncertain}) = \begin{cases} 1, & \alpha_t > \hat{\tau}_{1-\epsilon}, \\ 0, & \alpha_t \leq \hat{\tau}_{1-\epsilon}. \end{cases} \quad (7)$$

We define a caption-level nonconformity score as the average token score:

$$\bar{\alpha} = \frac{1}{T} \sum_{t=1}^T \alpha_t. \quad (8)$$

Using calibration caption scores

$$\mathcal{B}_{\text{cal}} = \{\bar{\alpha}^{(i)}\}_{i=1}^M, \quad (9)$$

we compute the caption-level threshold

$$\hat{\tau}_{1-\epsilon} = \text{Quantile}_{1-\epsilon}(\mathcal{B}_{\text{cal}}). \quad (10)$$

We further map token-level scores into four uncertainty bands (*low*, *medium*, *high*, *very high*) using calibrated thresholds corresponding to 68%, 95%, and 99.7% quantiles. After merging subword units into whole words, we obtain word-level uncertainty labels. A caption-level uncertainty label is computed analogously using $\hat{\tau}_q$.

This conformal framework preserves the predictive performance of the base model while introducing a principled and calibrated mechanism for uncertainty quantification. Unlike entropy-based or heuristic confidence measures, our method provides formal coverage guarantees and is compatible with any generative captioning model. The procedure is summarized in Algorithm 1.

Algorithm 1: Conformal Uncertainty-Aware Image Captioning

Require: Captioning model θ (BLIP-2 OPT6.7B), calibration set $\mathcal{D}_{\text{cal}}$, coverage levels $\mathcal{Q} = \{q_1, q_2, q_3\}$ (e.g., $q_1 = 0.68$, $q_2 = 0.95$, $q_3 = 0.997$)

- 1: **Calibration**
- 2: Initialize token score set $\mathcal{A}_{\text{cal}} \leftarrow \emptyset$, caption score set $\mathcal{B}_{\text{cal}} \leftarrow \emptyset$
- 3: **for** $(I^{(i)}, \hat{y}_{1:T}^{(i)}) \in \mathcal{D}_{\text{cal}}$ **do**
- 4: $s^{(i)} \leftarrow 0$ {caption-level accumulator}
- 5: **for** $t = 1$ to T **do**
- 6: $p_t^{(i)} \leftarrow p_\theta(\hat{y}_t^{(i)} | \hat{y}_{<t}^{(i)}, I^{(i)})$, $\alpha_t^{(i)} \leftarrow -\log p_t^{(i)}$ {token-level score}
- 7: $\mathcal{A}_{\text{cal}} \leftarrow \mathcal{A}_{\text{cal}} \cup \{\alpha_t^{(i)}\}$, $s^{(i)} \leftarrow s^{(i)} + \alpha_t^{(i)}$
- 8: **end for**
- 9: $\bar{\alpha}^{(i)} \leftarrow \frac{1}{T} s^{(i)}$, $\mathcal{B}_{\text{cal}} \leftarrow \mathcal{B}_{\text{cal}} \cup \{\bar{\alpha}^{(i)}\}$ {caption-level score}
- 10: **end for**
- 11: **for** $k = 1$ to 3 **do**
- 12: $\hat{\tau}_{q_k} \leftarrow \text{Quantile}_{q_k}(\mathcal{A}_{\text{cal}})$, $\hat{\tau}_{q_k} \leftarrow \text{Quantile}_{q_k}(\mathcal{B}_{\text{cal}})$ {thresholds}
- 13: **end for**
- 14: **Inference**
- Require:** Image I
- 15: Generate caption $\hat{y}_{1:T}$ with model θ
- 16: $s \leftarrow 0$ {caption-level accumulator}
- 17: **for** $t = 1$ to T **do**
- 18: $p_t \leftarrow p_\theta(\hat{y}_t | \hat{y}_{<t}, I)$, $\alpha_t \leftarrow -\log p_t$, $s \leftarrow s + \alpha_t$
- 19: **if** $\alpha_t \leq \hat{\tau}_{q_1}$ **then**
- 20: Label token $\hat{y}_t$ as *low* uncertainty
- 21: **else if** $\alpha_t \leq \hat{\tau}_{q_2}$ **then**
- 22: Label token $\hat{y}_t$ as *medium* uncertainty
- 23: **else if** $\alpha_t \leq \hat{\tau}_{q_3}$ **then**
- 24: Label token $\hat{y}_t$ as *high* uncertainty
- 25: **else**
- 26: Label token $\hat{y}_t$ as *very high* uncertainty
- 27: **end if**
- 28: **end for**
- 29: Merge subword units into whole words and aggregate token labels to obtain word-level uncertainty labels
- 30: $\bar{\alpha} \leftarrow \frac{1}{T} s$ {caption-level score}
- 31: **if** $\bar{\alpha} \leq \hat{\tau}_{q_1}$ **then**
- 32: Label caption as *low* uncertainty
- 33: **else if** $\bar{\alpha} \leq \hat{\tau}_{q_2}$ **then**
- 34: Label caption as *medium* uncertainty
- 35: **else if** $\bar{\alpha} \leq \hat{\tau}_{q_3}$ **then**
- 36: Label caption as *high* uncertainty
- 37: **else**
- 38: Label caption as *very high* uncertainty
- 39: **end if**
- 40: **Output:** generated caption $\hat{y}_{1:T}$ with word-level and caption-level uncertainty labels across four bands (*low*, *medium*, *high*, *very high*)

3.3. Theoretical analysis

The proposed framework combines a parametric image captioning model with a nonparametric calibration layer, forming a semi-parametric approach to uncertainty-aware caption generation. We distinguish between (i) distribution-free guarantees provided by conformal prediction and (ii) asymptotic properties arising from empirical estimation.

Validity of conformal calibration. Let α_{n+1} denote the nonconformity score of a new (test) sample drawn from the same distribution as the calibration data, and let $\hat{\tau}_{1-\epsilon}$ be the empirical quantile computed from the calibration set. Under exchangeability between calibration and test

samples, conformal prediction ensures

$$\mathbb{P}(\alpha_{n+1} \leq \hat{\tau}_{1-\epsilon}) \geq 1 - \epsilon. \quad (11)$$

This guarantee holds without assumptions on model correctness or the data distribution. In practice, exchangeability requires that calibration and test samples are drawn from the same underlying distribution. While the parametric model provides a ranking of predictions, the conformal calibration layer ensures valid uncertainty estimates that are independent of model correctness. This guarantee is marginal and does not imply conditional coverage across all subpopulations, where coverage may vary.

Consistency of empirical quantiles. Let $\{\alpha_i\}_{i=1}^n$ denote i.i.d. nonconformity scores computed on the calibration set, with empirical distribution $\hat{F}_n$ and population distribution F . Then

$$\sup_x |\hat{F}_n(x) - F(x)| \rightarrow 0, \quad (12)$$

and the empirical quantiles satisfy

$$\hat{\tau}_q \xrightarrow{p} \tau_q, \quad (13)$$

where τ_q denotes the corresponding population quantile and $\xrightarrow{p}$ denotes convergence in probability.

Thus, the estimated thresholds become increasingly accurate as the calibration size increases [41].

Model-induced scores. The nonconformity scores are defined as

$$\alpha_t = -\log p_\theta(\hat{y}_t | \hat{y}_{<t}, I). \quad (14)$$

Classical assumptions underlying M-estimation do not hold for modern neural networks. Therefore, the theoretical guarantees do not rely on consistency of the parametric model. Instead, the model provides a scoring function, while calibration ensures validity. In practice, numerical safeguards are applied to ensure stability for low-probability tokens.

Summary of guarantees.

- **Finite-sample validity:** Conformal calibration ensures marginal coverage under exchangeability.
- **Asymptotic accuracy:** Empirical quantiles converge to population quantiles.
- **Robustness to model misspecification:** Valid uncertainty estimates are maintained even if model probabilities are inaccurate.

Comparison with nonparametric methods. We compare our proposed conformal calibration framework with alternative nonparametric approaches, such as k -nearest neighbors (kNN) [42] and kernel density estimation (KDE) [43]. While kNN relies on local distances in feature space, it is highly sensitive to the choice of k and the underlying distance metric. In high-dimensional settings, kNN suffers from the curse of dimensionality and does not provide finite-sample coverage guarantees. Similarly, KDE estimates a smooth density, but its performance depends critically on bandwidth selection and degrades significantly in high-dimensional spaces. In contrast, our conformal calibration approach operates directly on scalar nonconformity scores, bypassing the need for high-dimensional density estimation. It provides distribution-free, finite-sample marginal coverage under the assumption of exchangeability, without requiring the tuning of sensitive parameters such as k or bandwidth. Consequently, the proposed framework offers a more robust and theoretically grounded solution for uncertainty estimation.

4. Experiments and results

In this section, we present our experimental setup and results on the ENVISAGE dataset and evaluate the conformal-based uncertainty-aware image captioning framework. We first evaluate baseline captioning models to establish baseline performance, then assess how

our uncertainty-aware approach enhances caption reliability and interpretability beyond the best-performing baseline.

4.1. Experimental setup

This subsection describes the setup used throughout our experiments. Both *Show and Tell* and *Show, Attend and Tell* baselines use a ResNet-50 encoder pretrained on ImageNet. *Show and Tell* produces fixed 2048-dimensional global image features, while *Show, Attend and Tell* extracts spatial feature maps of size $14 \times 14 \times 2048$ from the final convolutional block. The decoder in both is a four-layer LSTM with a hidden size of 1024, a word embedding dimensionality of 512, and a dropout rate of 0.5. Visual features are linearly projected to initialize the hidden and cell states of all LSTM layers, with *Show, Attend and Tell* using mean-pooled encoder features for initialization. In *Show, Attend and Tell*, the attention module projects encoder features and decoder hidden states into a shared 512-dimensional space. Training for both models employs teacher forcing and cross-entropy loss with padding tokens ignored, optimized using Adam with a learning rate of 1×10^{-4} , a batch size of 64, and a maximum of 200 epochs. Early stopping is applied based on validation performance with a patience of 15 epochs. Both *ViT-GPT2* and *CvT-GPT2* baselines employ a GPT-2 language decoder initialized from pretrained weights with cross-attention enabled. The tokenizer vocabulary is extended with special tokens (bos, eos, pad), and the embedding matrix is resized accordingly. For *ViT-GPT2*, visual features are extracted using a ViT-B/16 encoder pretrained on ImageNet, producing a fixed sequence of patch-level embeddings that are provided to the decoder as encoder hidden states with an all-ones attention mask. In contrast, *CvT-GPT2* uses a pretrained CvT-13 encoder, producing spatial feature maps of size $384 \times 14 \times 14$, which are flattened into patch tokens, linearly projected to the GPT-2 hidden dimensionality, and augmented with a global visual token computed as the mean of the patch embeddings. Training for both transformer-based models uses teacher forcing with a next-token prediction objective and cross-entropy loss while ignoring padded positions, optimized using AdamW with a learning rate of 5×10^{-5} . At inference time, captions are generated using greedy decoding. We fine-tune *BLIP* captioning models using the LAVIS framework. *BLIP Base* and *BLIP Large* use input images resized to 384×384 , while all *BLIP-2* variants (*BLIP-2 OPT2.7B*, *BLIP-2 OPT6.7B*, and *BLIP-2 T5*) use input images resized to 364×364 . All models are initialized from COCO-pretrained weights for fine-tuned evaluation on the ENVISAGE dataset. The default caption prompt (“a picture of”) is applied during both training and evaluation. Fine-tuning is performed for five epochs using a linear warmup followed by cosine learning-rate decay, with an initial learning rate of 1×10^{-5} and a weight decay of 0.05. *BLIP Base* uses batch sizes of 32 for training and 64 for evaluation, while *BLIP Large* uses batch sizes of 16 and 32, respectively, with gradient checkpointing enabled for memory efficiency. *BLIP-2* captioning models integrate a vision encoder with large language models via a query-based transformer and use 32 query tokens. All *BLIP-2* variants are trained with a batch size of 4 and evaluated with a batch size of 8. Training is conducted in distributed mode using two A6000 GPUs. Captions are generated using beam search with three beams, a minimum length of 5, and a maximum length of 20. Random seeds are fixed to ensure reproducibility. Uncertainty calibration is performed on the ENVISAGE validation set. Subword tokens are merged into word-level units using tokenizer boundary markers, and the maximum uncertainty level is propagated within each word. All generated captions are evaluated on the held-out ENVISAGE test set using standard COCO captioning metrics, including BLEU-1 through BLEU-4, METEOR, ROUGE-L, CIDEr, and SPICE.

Table 6

Performance comparison of classical, transformer-based, and pretrained captioning models on the ENVISAGE test set. *BLIP-2 OPT6.7B* (fine-tuned) achieves the best overall performance, while zero-shot models, including *GPT-4o*, lag behind fine-tuned baselines.

Model	BLEU-4	CIDEr	SPICE	Final Score
Non-pretrained baselines				
Show and Tell	0.1920	0.4417	0.1198	0.30944
Show Attend and Tell	0.1603	0.3495	0.1067	0.27344
ViT and GPT-2	0.1530	0.3686	0.1161	0.27856
CvT and GPT-2	0.1576	0.3654	0.1138	0.27696
Pretrained zero-shot models				
BLIP (Base Coco - Zero-Shot)	0.0880	0.3749	0.1270	0.25306
BLIP (Large Coco - Zero-Shot)	0.0937	0.3848	0.1265	0.25712
BLIP-2 OPT2.7b (Zero-Shot)	0.1045	0.4112	0.1319	0.26948
BLIP-2 OPT6.7b (Zero-Shot)	0.1074	0.4290	0.1340	0.27580
BLIP-2 T5 (Zero-Shot)	0.1044	0.4355	0.1397	0.27690
GPT-4o (Zero-Shot)	0.0443	0.3186	0.1366	0.21353
Fine-tuned models (on ENVISAGE)				
BLIP (Base Coco - Fine Tune)	0.3016	0.7823	0.1676	0.43486
BLIP (Large Coco - Fine Tune)	0.2946	0.7898	0.1717	0.43617
BLIP-2 OPT2.7b (Fine Tune)	0.2504	0.7095	0.1672	0.40433
BLIP-2 OPT6.7b (Fine Tune)	0.2956	0.7779	0.1699	0.43622
BLIP-2 T5 (Fine Tune)	0.2037	0.5366	0.1473	0.34946

4.2. Baseline performance

To establish a quantitative benchmark for ENVISAGE, we evaluate a diverse set of captioning models, including classical encoder-decoder networks, transformer-based architectures, and state-of-the-art pretrained models. Results on the test set show substantial performance differences across model families, guiding our selection of the best-performing baseline for subsequent uncertainty-aware captioning.

Table 6 shows the performance of classical, transformer-based, and pretrained captioning models on the ENVISAGE test set. Among the non-pretrained baselines, *Show and Tell* achieves the highest scores, with *Show, Attend and Tell* following closely. However, the overall performance of classical models remains limited. These architectures struggle to capture the complex object interactions, spatial relationships, and contextual cues prevalent in ENVISAGE, highlighting the need for stronger visual-linguistic representations.

Zero-shot pretrained models, including *BLIP*, *BLIP-2*, and *GPT-4o*, lag behind models fine-tuned on ENVISAGE. Despite strong results on datasets like MS COCO, these models struggle to generalize to the visually impaired perspective captured in ENVISAGE, which features off-center objects, unconventional viewpoints, cluttered indoor environments, and narrative requirements. This domain gap demonstrates that captioners trained on “photographic-style” datasets are not transferable to assistive scenarios without adaptation.

Fine-tuning yields substantial improvements across all pretrained models, particularly in metrics BLEU-4 and CIDEr, producing more descriptive and contextually accurate captions. Among all evaluated models, fine-tuned *BLIP-2 OPT6.7B* achieves the best performance on Final Score, demonstrating its superior ability to model the visual complexity and linguistic variability of ENVISAGE. Since captions for visually impaired individuals must be both correct and contextually informative, these findings justify two key decisions: (i) the necessity of dataset-specific fine-tuning over zero-shot deployment, and (ii) the selection of *BLIP-2 OPT6.7B* as the backbone for our uncertainty-aware captioning framework.

4.3. Uncertainty-aware captioning results

We qualitatively evaluate our conformal-based uncertainty framework on in-domain images from the ENVISAGE test set. Our goal is to assess whether the predicted uncertainty aligns with visual ambiguity



Generated Caption: artificial flowers are in the vase on the table
Seq. band: LOW **Avg NLL:** 6.615

(a)



Generated Caption: two teapots are on the table
Seq. band: MED **Avg NLL:** 7.878

(b)



Generated Caption: two people are playing in the water
Seq. band: HIGH **Avg NLL:** 8.799

(c)



Generated Caption: a market is photoed in the nighttime
Seq. band: VERY HIGH **Avg NLL:** 10.563

(d)

Fig. 1. Color-coded generated captions for four images from ENVISAGE with caption-level uncertainty bands and average negative log-likelihood (lower values indicate higher confidence). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 7

Captioning performance, distribution, and calibration error across uncertainty bands. (%) denotes the proportion of samples assigned to each band. Δ (%) indicates deviation from the expected Gaussian distribution (68, 27, 4.7, 0.3).

Method	Band	BLEU-4	CIDEr	SPICE	Final Score	(%)	Δ (%)
Ours	Low	0.2999	0.8124	0.1769	0.4476	68.31	+0.31
	Med	0.2924	0.7373	0.1593	0.4232	26.49	-0.51
	High	0.2569	0.6449	0.1347	0.3767	4.75	+0.05
	Very high	0.1861	0.4331	0.1045	0.3009	0.45	+0.15
KDE	Low	0.2957	0.7851	0.1738	0.4386	67.66	-0.34
	Med	0.2876	0.7996	0.1739	0.4399	27.39	+0.39
	High	0.2971	0.7702	0.1613	0.4308	4.35	-0.35
	Very high	0.2582	0.7003	0.1417	0.3876	0.61	+0.31
kNN	Low	0.2915	0.7769	0.1741	0.4354	46.27	-21.73
	Med	0.2974	0.8055	0.1743	0.4445	45.25	+18.25
	High	0.2873	0.7968	0.1629	0.4337	7.57	+2.87
	Very high	0.2401	0.6315	0.1466	0.3699	0.92	+0.62

and model confidence. The method produces meaningful uncertainty signals in most scenarios, highlighting challenging objects and occlusions. Table 7 presents captioning performance across four uncertainty bands for all methods. For the proposed method (Ours), a clear monotonic trend is observed: as caption-level uncertainty increases, performance consistently decreases across all metrics. The low uncertainty

band achieves the highest performance (Final Score = 0.4476), covering the majority of samples (68.31%), while scores progressively decline through the medium (0.4232) and high (0.3767) bands, reaching the lowest level in the very high uncertainty band (0.3009). This behavior is consistently reflected across BLEU-4, CIDEr, and SPICE, indicating that the proposed uncertainty measure is well-aligned with caption quality. In contrast, nonparametric baselines (KDE and kNN) do not exhibit a consistent monotonic relationship between uncertainty and performance. For KDE, the medium uncertainty band (Final Score = 0.4399) slightly outperforms the low uncertainty band (0.4386), and performance does not strictly decrease with increasing uncertainty. Similarly, kNN shows irregular behavior, with the medium band (0.4445) achieving higher performance than the low band (0.4354). These inconsistencies suggest weaker alignment between estimated uncertainty and actual caption quality for these methods. Overall, the proposed method not only achieves strong performance in the low uncertainty regime but also demonstrates a more reliable and interpretable ordering of samples according to uncertainty, which is critical for uncertainty-aware captioning systems.

Fig. 1 shows qualitative examples of uncertainty-aware captions generated by our framework on ENVISAGE. Each token is color-coded according to its uncertainty band: LOW, MEDIUM, HIGH, and VERY HIGH, as determined by the conformal quantile thresholds corresponding to approximately 68%, 95%, 99.7%, and beyond. The caption-level uncertainty band and average NLL are also reported to indicate overall model confidence.

In Fig. 1(a), the caption aligns well with the image, with uniformly low uncertainty across most tokens. The only notable exception is the word “vase”, which carries very high uncertainty—accurately reflecting model confusion about an object that is visually ambiguous. In Fig. 1(b), moderate uncertainty occurs for the word “teapots”, reflecting difficulty in identifying similar-looking objects, while the rest of the caption remains confident. In Fig. 1(c), the image shows people sitting or partially obscured under riverside pergolas rather than actively playing in the water. The medium uncertainty around the phrase “two people are” and high uncertainty on the verb “playing” correctly contribute to the overall HIGH caption-level uncertainty, as visual evidence for the described activity is ambiguous and does not clearly support the caption. Finally, the caption in Fig. 1(d) displays high uncertainty around multiple tokens (“is”, “photoed”, “nighttime”), leading to the overall VERY HIGH sequence band, suggesting low reliability and possible linguistic or visual ambiguity. These results demonstrate that token-level uncertainty identifies uncertain tokens, while caption-level bands provide an interpretable summary of overall confidence. Together, they reveal a strong correlation between uncertainty and caption quality: as uncertainty increases, captions tend to deviate from visually consistent descriptions. This underscores the value of uncertainty quantification for recognizing when the model *does not know*, a crucial capability for trustworthy assistive applications. Additional experimental results and dataset analyses are provided in the Supplementary Material.

5. Conclusion

In this work, we introduced ENVISAGE, a large-scale accessibility-focused image captioning dataset designed to support assistive vision-language systems for visually impaired individuals. ENVISAGE contains 53,452 images and 267,260 captions, with annotations emphasizing spatial relationships, contextual scene understanding, and task-relevant visual cues. Unlike general-purpose captioning datasets, ENVISAGE reflects real-world assistive scenarios in which captions must be not only linguistically accurate but also informative, concise, and reliable for user decision-making. We also proposed a conformal uncertainty-aware image captioning framework that augments a strong parametric captioning backbone with a nonparametric calibration layer. Through extensive benchmarking of classical encoder–decoder models, transformer-based captioners, and large pretrained vision-language models, we identified fine-tuned BLIP-2 OPT6.7B as the strongest backbone on ENVISAGE, achieving the best overall final score of 0.43622. Building on this model, the proposed conformal framework provides calibrated and interpretable uncertainty estimates at both token and caption levels. This enables the system to indicate not only what it predicts, but also how reliable each word and the complete caption are likely to be. The experimental results validate the effectiveness of the proposed framework from both performance and reliability perspectives.

Quantitative analysis shows that caption quality decreases monotonically as uncertainty increases: the low-uncertainty band achieves the highest final score of 0.4476, while the very-high-uncertainty band obtains the lowest score of 0.3009. This trend demonstrates that the estimated uncertainty is meaningfully aligned with caption quality and can serve as a practical reliability indicator. Qualitative examples further show that uncertain tokens often correspond to ambiguous objects, unsupported actions, or linguistically unreliable phrases, confirming the usefulness of token-level uncertainty for interpreting model behavior. The proposed framework therefore contributes to more reliable assistive image captioning by combining dataset-specific caption generation with statistically grounded uncertainty calibration. In practical assistive settings, this is important because incorrect but confident captions may reduce user trust or lead to unsafe interpretation of the surrounding environment. By explicitly communicating uncertainty, the system can support more informed user interaction, enable confidence-aware

downstream decisions, and provide a foundation for safer deployment of vision-language models in accessibility applications.

Despite these contributions, several limitations remain. First, ENVISAGE was mainly collected from visually impaired individuals in the İzmir region of Türkiye, and may therefore reflect region-specific environments, cultural patterns, and scene distributions. Second, although conformal prediction provides distribution-free marginal guarantees under exchangeability, performance may degrade under strong domain shift, such as images collected in different countries, indoor environments, or sensor conditions. Third, the current framework identifies uncertainty but does not yet actively revise captions or request additional user input when uncertainty is high. Future work will address these limitations in several directions. We plan to expand ENVISAGE with more geographically and culturally diverse data, conduct user studies with visually impaired individuals to assess real-world usefulness, and investigate interactive human-in-the-loop caption refinement for uncertain predictions. Further work will also explore uncertainty-guided decoding, caption reranking, privacy-preserving deployment, and hybrid uncertainty models that combine conformal calibration with model-based uncertainty estimates. These directions will help advance reliable, transparent, and user-centered assistive vision-language systems for real-world use.

CRedit authorship contribution statement

Özkan Çaylı: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation. **Volkan Kılıç:** Writing – review & editing, Writing – original draft, Supervision, Data curation, Conceptualization. **Wenwu Wang:** Writing – review & editing, Writing – original draft, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.patcog.2026.114298>.

Data availability

The dataset is publicly available at: https://github.com/cayli-crypto/envisage_dataset.

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